

Natural Computing

LMU Munich
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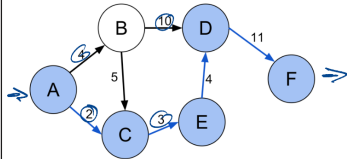


Definition 9 (optimization). Let $\mathcal{X}$ be an arbitrary state space. Let $\mathcal{T}$ be an arbitrary set called target space and let $\leq$ be a total order on $\mathcal{T}$. A total function $\tau : \mathcal{X} \rightarrow \mathcal{T}$ is called target function. Optimization (minimization/maximization) is the procedure of searching for an $x \in \mathcal{X}$ so that $\tau(x)$ is optimal (minimal/maximal). Unless stated otherwise, we assume minimization. An optimization run of length $g + 1$ is a sequence of states $\langle x_t \rangle_{0 \leq t \leq g}$ with $x_t \in \mathcal{X}$ for all t .

often: $\mathcal{T} = \mathbb{R}$

$\langle x_0, \dots, x_g \rangle$

Shortest Path



↳ example

$$x = \{ \underline{(A,C)}, \underline{(C,E)}, \underline{(E,D)}, \underline{(D,F)} \}$$

$$\mathcal{X} = \mathcal{P}(E) = 2^E$$

for graph edges E

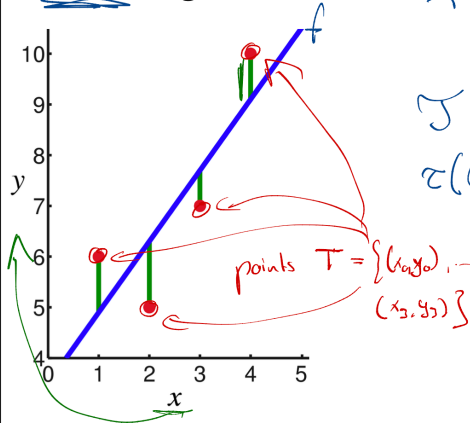
$$\mathcal{Y} = \mathbb{Z}$$

$$\tau = \{e_1, \dots, e_n\} \mapsto \sum_{i=1}^n \text{cost}(e_i)$$

$$\tau: \mathcal{X} \rightarrow \mathcal{Y}$$

$$x^* = \underset{x \in \mathcal{X}}{\text{argmin}} \tau(x)$$

Linear Regression



$$\mathcal{X} = \mathbb{R}^2$$

so that $f(x) = \underline{ax} + \underline{b}$ for $\underline{x} = (x, 1)$

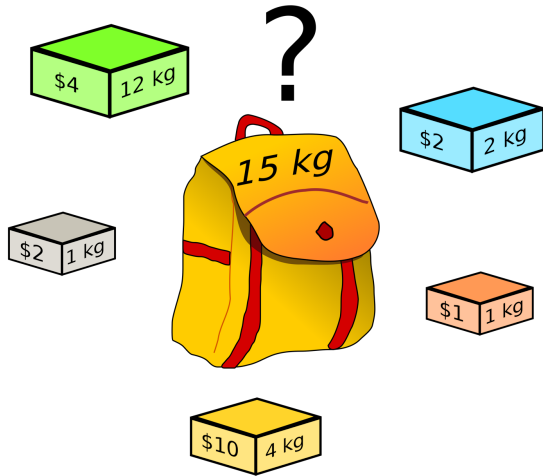
$$\mathcal{Y} = \mathbb{R}$$

$$\tau(a, b) = \sum_{i=0}^{n-1} |(ax_i + b) - y_i|$$

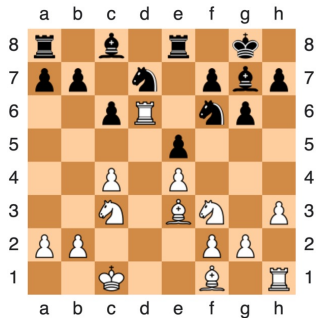
$$\mathcal{F} = \{f: \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is linear}\}$$

↳ mathematically possible
but very inefficient

The Knapsack Problem

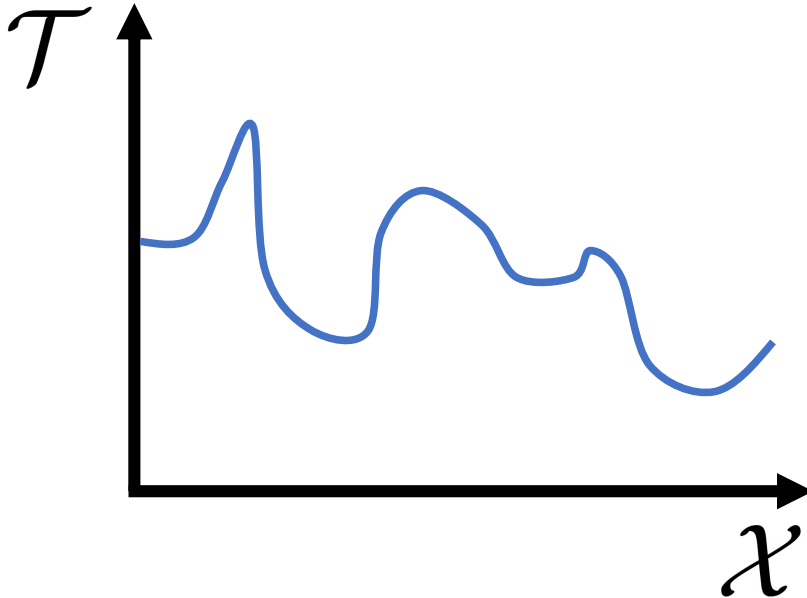


Playing Chess

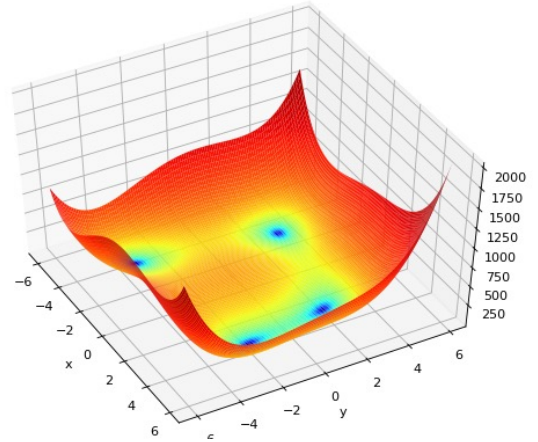
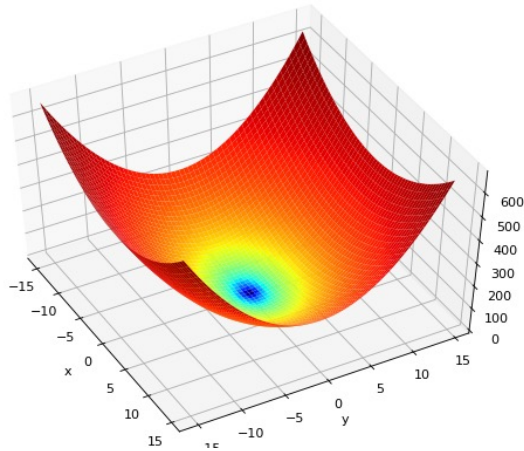


Further Examples

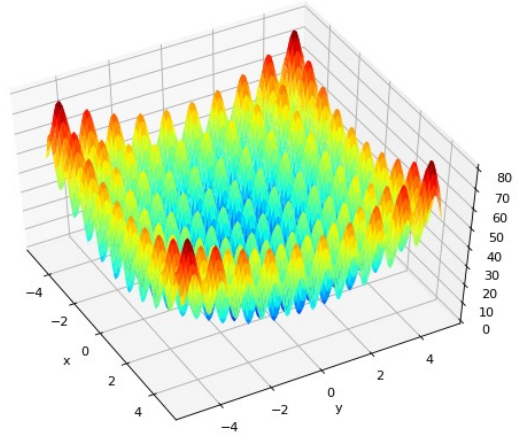
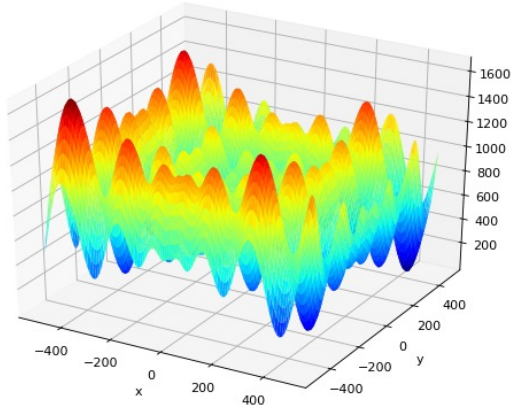
The Solution Landscape Metaphor



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We use the following terminology:

- An optimization run is *working to some extent* if $t \leq t'$ for $t, t' \in [1; g] \subset \mathbb{N}$ correlates positively to some extent with $\tau(x_t) \geq \tau(x_{t'})$.
- An optimization run is *improving* if $\tau(x_t) \geq \tau(x_{t'})$ for all $t \leq t'$.
- An optimization run has *found a global optimum* iff $\tau(x) \geq \tau(x_g)$ for all $x \in \mathcal{X}$.

Let $e : \langle \mathcal{X} \rangle \times (\mathcal{X} \rightarrow \mathcal{T}) \rightarrow \mathcal{X}$ be a possibly randomized or non-deterministic function so that the optimization run $\langle x_t \rangle_{0 \leq t \leq g}$ is produced by calling e repeatedly, i.e., $x_{t+1} = e(\langle x_u \rangle_{0 \leq u \leq t}, \tau)$ for all t , $1 \leq t \leq g$, where x_0 is given externally (e.g., $x_0 =_{\text{def}} 42$) or chosen randomly (e.g., $x_0 \sim \mathcal{X}$). An optimization process is a tuple $(\mathcal{X}, \mathcal{T}, \tau, e, \langle x_t \rangle_{0 \leq t \leq g})$.

We use the following terminology:

- The *end result* of an optimization process $\mathcal{D} = (\mathcal{X}, \mathcal{T}, \tau, e, \langle x_t \rangle_{0 \leq t \leq g})$ is x_g .
- The *best result* of an optimization process $\mathcal{D} = (\mathcal{X}, \mathcal{T}, \tau, e, \langle x_t \rangle_{0 \leq t \leq g})$ is

$$*\mathcal{D} = \arg \min_{\substack{x_t \\ 0 \leq t \leq g}} \tau(x_t)$$

with the *best target value* $|\mathcal{D}| = \tau(*\mathcal{D})$.

- An optimization process $\mathcal{D}' = (\mathcal{X}, \mathcal{T}, \tau, e, \langle x'_t \rangle_{0 \leq t \leq g'})$ is a *continuation* of an optimization process $\mathcal{D} = (\mathcal{X}, \mathcal{T}, \tau, e, \langle x_t \rangle_{0 \leq t \leq g})$ iff $g' \geq g$ and $x'_t = x_t$ for all t with $0 \leq t \leq g$. We write $\ell(\mathcal{D})$ for the set of all possible continuations of $\mathcal{D}$, which can be large if e is randomized or non-deterministic.
- An optimization process $\mathcal{D}$ is *elitist* iff for all its continuations $\mathcal{D}' = (\mathcal{X}, \mathcal{T}, \tau, e, \langle x'_t \rangle_{0 \leq t \leq g'}) \in \ell(\mathcal{D})$ it holds that $\tau(x'_t) \geq \tau(x'_{t'})$ for all $t \leq t'$.

How do we optimize for τ ?

Algorithm 1 (single-sample random search). Let $\mathcal{D} = (\mathcal{X}, \mathcal{T}, \tau, e, \langle x_u \rangle_{0 \leq u \leq t})$ be an optimization process. The process $\mathcal{D}$ continues via (single-sample) random search if e is of the form

$$e(\langle x_u \rangle_{0 \leq u \leq t}, \tau) = x_{t+1} = \arg \min_{x \in \{x_t, x'_t\}} \tau(x)$$

where $x'_t \sim \mathcal{X}$ is drawn at random for each call to e .

Algorithm 2 (stochastic hill climbing). Let $\mathcal{D} = (\mathcal{X}, \mathcal{T}, \tau, e, \langle x_u \rangle_{0 \leq u \leq t})$ be an optimization process. Let $neighbors : \mathcal{X} \rightarrow \wp(\mathcal{X})$ be a function that returns a set of neighbors of a given state $x \in \mathcal{X}$. The process $\mathcal{D}$ continues via stochastic hill climbing if e is of the form

$$e(\langle x_u \rangle_{0 \leq u \leq t}, \tau) = x_{t+1} = \arg \min_{x \in \{x_t, x'_t\}} \tau(x)$$

where $x'_t \sim neighbors(x_t)$ is drawn at random for each call to e .

Algorithm 3 (simulated annealing). Let $\mathcal{D} = (\mathcal{X}, \mathcal{T}, \tau, e, \langle x_u \rangle_{0 \leq u \leq t})$ be an optimization process. Let $neighbors : \mathcal{X} \rightarrow \wp(\mathcal{X})$ be a function that returns a set of neighbors of a given state $x \in \mathcal{X}$. Let $\kappa : \mathbb{N} \rightarrow \mathbb{R}$ be a temperature schedule, i.e., a function that returns a temperature value for each time step. Let $A : \mathcal{T} \times \mathcal{T} \times \mathbb{R} \rightarrow \mathbb{P}$ with $\mathbb{P} = [0; 1] \subset \mathbb{R}$ be a function that returns an acceptance probability given two target values and a temperature. Commonly, we use

$$A(Q, Q', K) = e^{\frac{-(Q' - Q)}{K}}$$

for $\mathcal{T} \subseteq \mathbb{R}$ and Euler's number e . The process $\mathcal{D}$ continues via simulated annealing if e is of the form

$$e(\langle x_u \rangle_{0 \leq u \leq t}, \tau) = x_{t+1} = \begin{cases} x'_t & \text{if } \tau(x'_t) \leq \tau(x_t) \text{ or } r \leq A(\tau(x_t), \tau(x'_t), \kappa(t)), \\ x_t & \text{otherwise,} \end{cases}$$

where $x'_t \sim neighbors(x_t)$ and $r \sim \mathbb{P}$ are drawn at random for each call to e .